Occupational health

PROF. DR. FOUAD METRY ATIA

PROFESSOR OF COMMUNITY MEDICINE

Intended learning outcome

- By the end of the lecture the student will be able to :
- Identify concept and aim of occupational medicine
- Identify different health effects from different hazards
- Understand clinical features and management of these health effects
- Outline Important items of prevention and control programs for occupational health problems

Definition of occupational health :

- World Health Organization (WHO), define occupational health as the promotion and maintenance of the highest degree of physical, mental and social well-being of workers in all occupations.

Aim of occupational health

- To increase the efficiency
- To increase the production
- To decrease the accidents

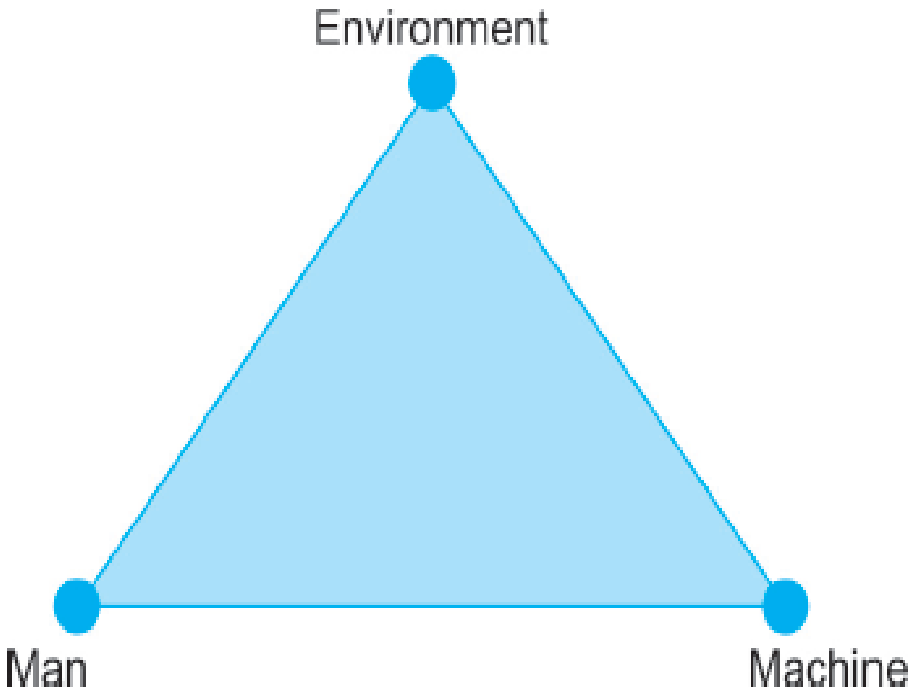
Objectives of occupational health

- To promote the health of the workers
- To maintain the highest degree of physical, mental and social well-being of the workers
- To prevent the diseases by elimination of factors which are inimical to their health.

Working (Occupational) Environment

- It is the environmental conditions prevailing in the working place, which have a bearing or influence on the health of the worker, by three types of interactions. In othe words there are three types of working environment .

- First one, ‘man with machine
- Second one, ‘man with environment
- Third one, ‘man with man’:



Role of Occupational Medicine

Most Common Activities of Occupational Medical Physicians Worldwide

1. Occupational health-risk/targeted screenings
2. Traditional occupational injury/illness management
3. Primary health care
4. Return-to-work examinations
5. Pre-employment examinations
6. Administrative duties
7. General health screenings
8. Workers' compensation
9. Health promotion/disease prevention
10. Evaluation of worksite environment
11. Health education
12. Disability management
13. Training/teaching/mentoring
14. Research
15. Emergency medical care
16. Safety assessments of workplace
17. Health benefits management
18. Evaluation of effects of worksite activities in external environment

Ergonomics

- ‘Ergonomics’ is derived from two Greek words ‘Ergon’ meaning work and ‘Nomos’ meaning principles or laws.
- Ergonomics is the Science of Work.
- Ergonomics means **placing** the worker in an environment (job), which is adapted to his physiological and psychological capacity.
- Ergonomics is essentially **fitting** the workplace to the worker. The better the fit the higher the level of safety and worker efficiency.
- The main object of ergonomics is to achieve the best **mutual adjustment** between the man and the machine, which are complimentary to each other.
- It takes account of the worker’s **capabilities and limitations** in seeking to ensure that tasks, equipment, information and the environment suit each worker.

The benefits of good Ergonomics are

- Improved health and **safety** by reducing work – related injuries and disorders;
- Improved **comfort**, morale and job **satisfaction**;
- Improved **productivity** and reduced workers' compensation costs and employee turnover.

Occupational health hazards

- Depending upon the nature of the occupation, an industrial worker is exposed to one or more of the following five types of hazards
  - Physical,
  - Chemical,
  - Biological,
  - Mechanical and
  - Psychosocial hazards),

Physical hazards

Health hazards due to exposure to heat: The effects are

- **Heat syncope:** It is fainting due to pooling of blood in lower limbs.
- **Heat cramps:** It is painful and spasmodic contractions of muscles due to loss of sodium and chloride
- **Heat exhaustion:** It means loss of salt leading on to circulatory failure.

- **Heat hyperpyrexia:** It is characterized by failure in heat regulating mechanism without the features of heat stroke. Temperature is about 106°F (40 °C). It may proceed to heat stroke.
- **Heat stroke:** There will be failure in the heat regulating mechanism, resulting in high temperature of the body, delirium, convulsions, partial or total loss of consciousness. Skin is dry and hot.
- **Prickly heat**
- **Burns**, etc.

Health hazards due to exposure to cold: The effects are

- Frost bite
- Reynaud's disease
- Erythromelalgia
- erythrocyanosis,
- chilblains
- trench-foot,
- gangrene, etc.

  - All because of cutaneous vasoconstriction.

Health hazards due to exposure to Light: The effects are

- Effects occur either due to inadequate light or excessive light.
- **Inadequate light:**
  - Headache, eye strain, eye fatigue (visual fatigue).
  - Chronic effect is ‘Miner’s nystagmus’.
- **Excessive light:**
  - Bright light results in glaring, visual fatigue, blurring of vision, discomfort and accidents.

Health hazards due to exposure to Radiation: The effects are

- **1. Ionizing radiation:** X-rays, radio-active isotopes like cobalt 60, phosphorus 32.
  - **i. Acute effects:** Acute radiation syndrome .
  - **ii. Chronic effects:**
    - **Somatic effects:** Leukemia, aplastic anemia, cancer, tumor induction, pancytopenia.
    - **Genetic effects:** Stillbirths, congenital defects, neonatal deaths, sex chromosome aneuploidy, sterility

2. Nonionizing radiation:

- **i. Ultraviolet radiation:**
  - On the skin: Darkening of the skin, thickening of the skin, erythema, cancer of the skin.
  - On the eyes (Ex: Welding): Photophobia, conjunctivitis, keratitis, corneal ulcer, blindness (Welder's flash), snow blindness.
- **ii. Infrared radiation:** Cataract
- **iii. Microwaves:** Microwave sickness
- **iv. Laser radiations:** Thermal burns, cataract, retinal burns.

Health hazards due to exposure to atmospheric pressure: The effects are

i. High atmospheric pressure (as in mines):

- Caisson disease (compressed air illness), syncope, aches and pains in joints, air embolism, cardio-respiratory distress, paralysis.

ii. Low atmospheric pressure (as in high attitudes):

- Deafness (due to blocking of Eustachian tubes),
- pain in the ears,
- rupture of eardrums,
- expansion of gasses in the sinuses and body cavities,
- headache,
- pulmonary edema,
- dyspnea, etc.

Health hazards due to exposure to Electricity: The effects are

- Electric shock,
- burns,
- ventricular flutter.

Health hazards due to exposure to vibration (working with pneumatic tools like drills and hammers): The effects are

- Numbness,
- white fingers,
- injuries.

Health hazards due to exposure to Noise: The effects are

Auditory effects:

- Deafness (temporary or permanent),
- tinnitus (buzzing in the ears)
- degeneration of cochlea and eighth nerve.

Nonauditory effects:

- Fatigue,
- irritability,
- nervousness,
- interference with speech and communication,
- annoyance,
- increased intracranial tension,
- hypertension,
- peptic ulcer,
- higher environmental stress.

Chemical hazards: These result from

Inhalants: Dusts, fumes and gases.

- **Dusts**: Pneumoconioses.
- **Fumes**: Metal fume fever (This chemical intoxication results from inhalation of fumes of molten metals like arsenic, antimony, beryllium, cadmium, cobalt, lead, zinc, etc.)
- **Gasses**:
  - **Asphyxiating gasses**: CO, SO2, Cl2, H2S, methyl isocyanide gas, etc.
  - **Anesthetic gasses**: Ether, chloroform, trichloroethylene.

- **Irritants** (Such as dyes, acids and alkalis): Occupational dermatoses, e.g. dermatitis, folliculitis, eczema, ulcerations, cancer.
- **Ingestants**: Toxic hazards occurring from the metals like lead, arsenic, mercury, cadmium, manganese, chromium, etc.
- **Allergens**: Allergic rhinitis, bronchitis, asthma, dermatitis, urticaria, etc.

Biological hazards

These are from the animals and soil. These are common in agricultural industry.

- **From the animals:**
  - They are called 'Zoonotic diseases'. E.g., Anthrax, bovine tuberculosis, salmonellosis, Japanese encephalitis, rabies, plague, Kyasanur forest disease, etc.
- **From the soil:**
  - Tetanus, ankylostomiasis, gas-gangrene, malignant edema, anthrax, aspergillosis, coccidioidomycosis, mycetoma.

Mechanical hazards

- These are mainly accidents.

Psychosocial hazards

These are due to failure of the worker to develop a healthy relationship with his co-workers, employers, management, supervisors, etc.

They are divided into two groups.

- **Psychological (Behavioral) changes:**
  - Such as hostility, aggressiveness, anxiety, depression, frustration, tardiness, alcoholism, drug addiction, sickness absenteeism, etc.
- **Psychosomatic ill health:**
  - Such as neurosis, fatigue, propensity to peptic ulcer, hypertension, asthma, etc.

Other Hazards

- **Occupational cancers:**
  - Cancers of the skin, lungs, bladder and blood forming organs (i.e. Leukemia).
- **Occupational dermatoses:**
  - Dermatitis, eczema, folliculitis, urticaria, ulcers, etc.

Definition of Occupational diseases

Occupational diseases occur as a result of exposure to physical, chemical, biological or psychosocial factors in the workplace.

Occupational diseases are

- cause-specific,
- occurring mainly among working population,
- exposure at workplace is essential,
- notifiable and
- compensable.

- However, some other diseases where work conditions or stress can play a role in their occurrence, these are called **work-related.**
- These diseases are also frequently seen in the general community.
- In these diseases, work may be a factor in its causation or may aggravate a pre-existing condition.

- **Examples of work-related diseases are:**
  - Hypertension.
  - Ischemic heart disease.
  - Psychosomatic illness.
  - Musculoskeletal disorders.

PNEUMOCONIOSES

(Pneumons = lungs; Konia = dust; Pneumoconiosis is singular).

- These are also known as dust diseases.
- Pneumoconioses are a group of lung diseases occurring out of the specific occupation, caused by inhalation of insoluble dust, over a prolonged period of exposure.
- Pathologically it is characterized by fibrosis of the lung parenchyma followed by its complications.
- Once a pathology, it is usually progressive, permanent, pulmonary pathology.

- **Clinically**, it is characterized by persistent **cough**, progressive **breathlessness**, gradually cripples the person by reducing the working capacity due to fibrosis of the lungs followed by the features of **complications** such as tuberculosis, emphysema, chronic obstructive pulmonary disease (COPD), pulmonary hypertension, cor pulmonale and even carcinoma in some cases.
- There is **no treatment or cure** for pneumoconioses.
- **Prevention is the only intervention**. Therefore, it is essential to prevent these diseases from arising.

Factors influencing pneumoconioses are:

- **Concentration of the dust in the air:**
  - Higher the concentration, greater the health hazards. The **permissible** limit is 200 mcg per cu meter of air.
- **Composition of the dust:**
  - More **complicated** the composition of the dust, **greater** the health hazard.

Size of the dust particles:

- This is the most important determining factor. **Smaller** the size of the dust particle, greater the tissue reaction.
  - Particles more than **10** µm, **settle** down on the ground and become soil dust.
  - Between **10 and 5** µm, the particles are **caught** in the **upper** respiratory passage.

- **Between 5 and 3 µm, they are caught in mid respiratory passage.**
- **Between 3 and 0.5 µm, they reach the alveoli and cause tissue reaction (fibrosis). They are most dangerous.**
- **Less than 0.1 µm, they are harmless because of Brownian movements they exhibit and they are absorbed.**

- **Duration of exposure:**
  - **Longer** the duration of exposure to the dust, **greater** the health hazard.
  - Nearly **10 to 15** years of exposure is necessary for the development of tissue reaction.
  - However, there are cases on record in which extensive fibrosis has occurred within **2** years following exposure to **high** concentration of dust.
- **Individual susceptibility (Health status):**
  - **Better** the health and nutritional status, **lesser** the chances of development of pneumoconiosis early.

SILICOSIS

Silicosis is a nodular fibrosis of the lung due to inhalation of respirable dust containing free silica (Silicon dioxide).

- **Exposure: Workers in the following industries are exposed to silica dust:**
  - **Mining.**
  - **Glass industry.**
  - **Pottery industry.**
  - **Ceramic industry.**
  - **Sandblasting and mineral polishing.**
  - **Foundry workers.**

Pathogenesis

- The inhaled dust is engulfed by macrophages.
- Because the silica is cytotoxic, the macrophages rupture and eliberate fibroblast attractive substances.
- This initiate fibrosis and formation of fine nodules that may coalesce together to from larger nodules.
- The hilar lymph nodes are enlarged with calcium deposition at the periphery giving the characteristic picture of **eggshell calcification or snowstorm appearance** in the X Ray .
- Egg shell appearance may also occurs in sarcoidosis and Tuberculosis.

Diagnosis of silicosis:

1-Occupational history: working in jobs entailing inhalation of silica dust for a period of 5-10 years.

2- clinical examination demonstrates symptoms of lung fibrosis e.g. shortness of breath, dyspnea, easily fatigability.....etc cough, expectoration may be present in case of concomitant chronic bronchitis.

Local chest expectoration is not specific.

3- X ray of the lung shows nodular opacities of varying size more at the upper parts of the lungs. There may be hilar eggshell calcification or snowstorm appearance.

4- Ventilatory function tests show drop in forced vital capacity.

Complications:

1- Silicosis is a risk factor for **TB** infection

2- Progress to chronic obstructive pulmonary disease (**COPD**), emphysema, pulmonary hypertension, cor-pulmonale

3- Recent studies point to the relation between silica exposure and **bronchogenic carcinoma**.

4- **Coplan's syndrome** (i.e. Silicosis with rheumatoid arthritis).

Prevention

Follow the general rules. Note that:

1- Environmental measures are the most important line of prevention:

- **Substitution of sand by coundum (Al2o3) as an artificial abrasive for sand blasting.**
- **Wetting the dust during preparation and shaping.**

2- Periodic medical examination is by clinical examination and x ray each two years.

Asbestosis

- Asbestos can induce different diseases in different sites of the body.
- So, it is called now Asbestos related diseases.
- Asbestos is imported from Canada, south Africa, Mexico and brazil

Occupational Exposure and uses in Egypt:

- Asbestos is a unique natural fibrous material with the following advantages:
  - Very **resistant to friction** so used in the manufacture of car brakes and clutch.
  - Highly **insulating** so used in clothes and gloves for over workers and for insulation of roofs.
  - Have a **binding** capacity so used in asbestos cement industry for manufacture of water pipes, ceiling tiles & roofs and walls.

Pathogenesis

- The fibers are cytotoxic to the macrophages.
- Rupture of the macrophages, liberate fibro-genic materials which induce the following:
- Diffuse fibrosis of the lung due to exposure to asbestos. The fibers may be surrounded by ferritin granules giving the so-called asbestos bodies. These are yellow brown in color, beaded and clubbed ended. They can be detected **microscopically in the sputum** of asbestos exposed workers. They are indicators of exposure to asbestos but not diagnose asbestosis.
- Interstitial fibrosis especially at the lower half of the lung.

- **X-ray chest shows**
  - **‘ground-glass’ or (honey – comb) appearance of lower two thirds of the lungs and heart shows ‘shaggy’ appearance (i.e. without clear-cut cardiac margins).**
  - **Pleural thickening**
  - **Pleural plaque: Cartilaginous thickening in the parietal pleura**
  - **Pleural effusion (rare)**

Diagnosis of asbestosis:

- 1- Occupational history:
  - history of exposure to asbestos fibers for a period of 5-10 years.
- 2- Clinical examination
  - demonstrates symptoms of lung fibrosis e.g Shortness of breath, dyspnea, easily fatigability.. etc
  - Osculation may reveal fine crepitations at the base of the lung due to fibrosis.

- **3- X ray shows:**
  - **‘ground-glass’ or (honey – comb) appearance of lower two thirds of the lungs and heart shows ‘shaggy’ appearance, tenting of the diaphragm, Pleural thickening, pleural effusion, pleural plaques.**
- **4- Pulmonary function tests show**
  - **drop in forced vital capacity**
- **5- Sputum examination may reveal asbestos bodies.**
  - **Other asbestos related diseases cancer larynx, and chronic Asthmatic bronchitis**

Complications:

- **Neoplasms**: Common malignant lesion is **bronchogenic** carcinoma. Less common is malignant **mesothelioma** of pleura or peritoneum, often associated with effusion or ascites respectively
- Cor-pulmonale.
- Asthmatic bronchitis is common especially among workers of cement asbestos pipes.

Prevention:

- The same lines of prevention applied with silicosis are applied here with workers exposed to asbestosis.

Bagassosis

- Extrinsic allergic alveolitis
  - due exposure to spores of the fungi thermoactinomyces sacchari found on the stored bagass in the wood and paper industry.
- The pathological process
  - begins by diseases inflammatory process in the form of granulomatous pneumonitis that occasionally undergoes organization leading to
  - interstitial fibrosis and thickening of the alveolo-capillary membrane.

Clinical presentation

it is in the form of acute phase or chronic phase

- **Acute form:**
  - **Onset: it occurs 4-6 hours after exposure.**
  - **Symptoms:**
    - **The symptoms are in the form of malaise, fever, chills, pains, unproductive cough, chest tightness and dyspnoea.**
    - **These symptoms last from 12 hours up to a week.**
  - **Signs:**
    - **fever, tachycardia and tachypnoea and basal crepitations.**

- **Chronic form:**
  - **show clinical features as in other pneumoconiosis.**

- **Investigations:**
  - **X-ray of the chest shows 'mottling' appearance in lung fields**
  - **Serology, pulmonary function tests, and lung biopsy.**

Prevention:

- **The most important is the prevention of growth of the fungi on the hay or the bagasse by spraying 1% propionic acid. Otherwise follow the general rules.**

Byssinosis:

- Asthma like manifestation occurs among workers exposed to cotton dust in textile industry in the wars of bale opening, carding combing, spinning and sometimes weaving.
- Although cotton industry is wide spread in Egypt, This disease is rare.

Diagnosis :

- history of exposure to cotton dust to more than 10 years.
- The worker suffers of breathlessness and chest tightness and sometimes cough on the day after a day off.
- Clinical examination: Picture of chronic obstructive airways disease.

- X ray is usually **normal** except some increase in bronchovascular markings.
- Pulmonary function tests show drop in forced expiratory volume especially at the shift end of the first working day after a holiday.

➢ **Treatment**:

- as in asthma or asthmatic bronchitis.

➢ **Prevention**:

- Follow general rules.

Mill fever:

- Occurs in new employees unaccustomed to dust although tolerance develops after few days of acclimatization.
- Symptoms:
  - Acute onset of fever, chills, cough and general malaise.
- They start after 5-6 hours from exposure and clear rapidly on removal of patient from the dusty environment.

Weaver's cough:

- Occurs in workers exposed to low grade wet cotton containing moulds and fungi.
- It affects all workers, old and new in an epidemic form, after 4-6 hours from exposure.
- Symptoms:
  - Mild dyspnea and persistent irritating cough with yellowish green sputum,
  - acking limbs and back,
  - headache, slight evening temperature and
  - epistaxis in some cases.

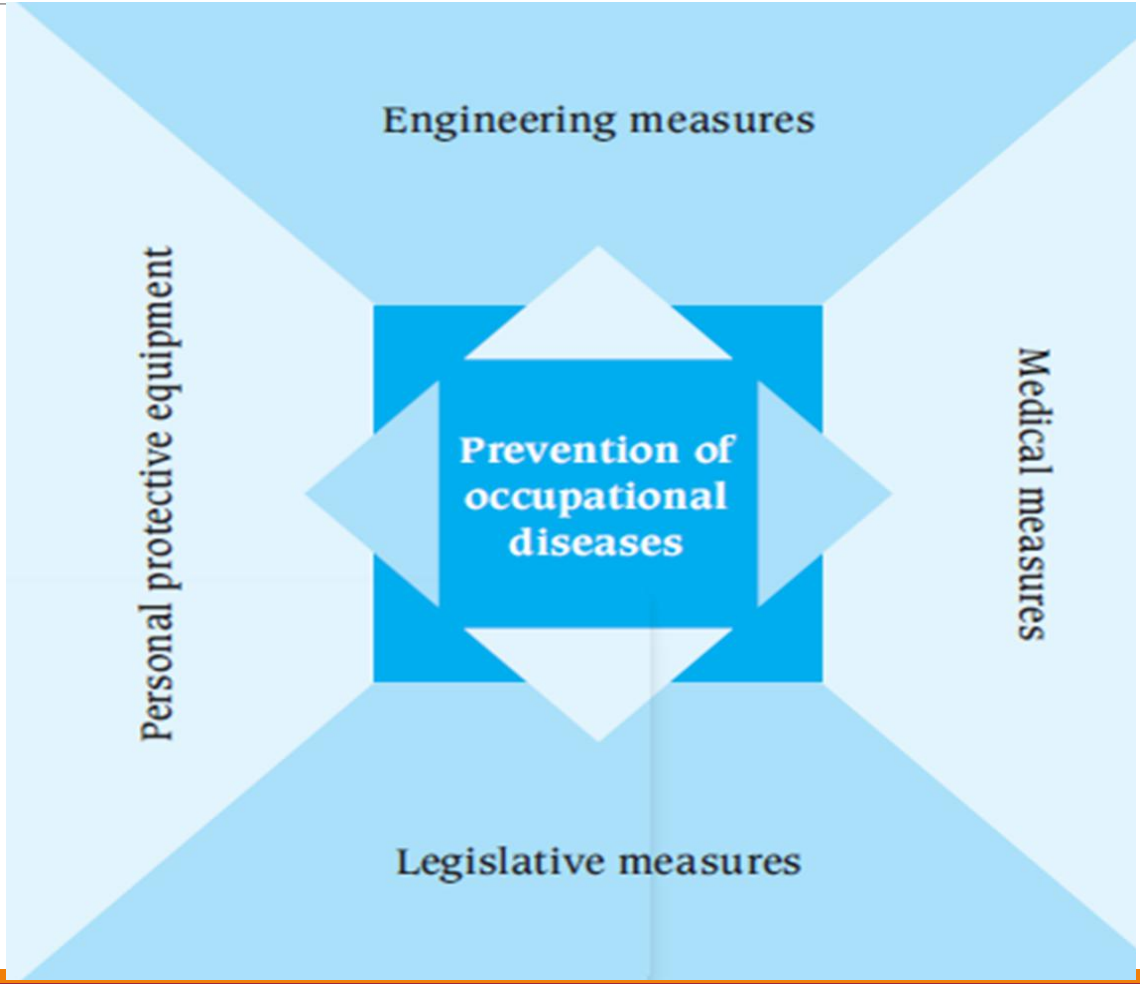
Farmer's lung

- This is a chronic disease of lungs, due to inhalation of **mouldy hay or grain dust**, in the agricultural fields.
- During winter season, the grain-dust with a moisture content of over 30 percent, bacteriae grow rapidly and produce heat of about 45°C.
- The heat encourages the growth of certain thermophilic actinomycete fungi of which Microspora (Micropolyspora) faeni is the main cause of Farmer's lung.

- **This condition is considered to be an allergic (hypersensitivity) reaction to the inhalation of antigen (dust) with the development of antibodies.**
- **The clinical features are the same as that of bronchial asthma (Eosinophilia) (Allergic bronchitis). First acute attacks are noticed.**
- **Repeated attacks cause pulmonary fibrosis and lung damage (cor pulmonale).**
- **X-ray chest shows 'fine nodular density'.**
- **Since agricultural industry is the major industry in our country, this is one of the major public health problems of our country**

General Measures of Prevention & Control of Occupational Diseases

Measures of Prevention and Control of occupational diseases are shown in the figure



1. Engineering measures

- Engineering methods for prevention of occupational hazards are:

- **Substitution** : Substitution of hazardous substances or operation by non-hazardous ones or e.g.,
  - fiberglass in place of asbestos,
  - compression instead of hammering.
- **Isolation**:
  - isolation of the hazardous process inside the work places so that other surrounding workers are not affected e.g. isolation of radiation or noise sources.
- **Segregation:**
  - segregation of the hazardous process from the work place. e.g. incinerators.

- **Good housekeeping :** The golden rule is that there is a place for everything and everything has a place.
- **Good general ventilation :** It dilutes the substance in work environment.
- **Periodic environmental surveys** : These will help in detection of any leak at an early stage. Threshold limit value of a substance should not be exceeded.
- **Safe plant design :** Safety engineers should be involved in the construction of the plant. A positive approach has been found to be more effective.

2. Medical Control Measures

➢ Medical examination

- **Pre placement medical examination :**
  - It is done before one is employed in a particular job i.e. at the time of entry into an employment and
  - includes a detailed history followed by complete physical examination and laboratory/ radiological investigations of the individual.
  - It helps in rejecting the individuals who are unfit and also helps in placement of the worker to the right job so that there is maximum productivity and at the same time it is less detriment to worker's health.
  - It also gives a baseline for various parameters which can be used for future comparisons.

- **Periodical medical examination:**
  - Early detection of health effects of hazardous exposure to be treated in its reversible stage.
  - Monitoring the efficiency of occupational heath services.

- **Health Education**
  - Health education about the process, handling of raw material, correct use of protective devices has a lot of impact in prevention of occupational diseases.
  - The aim is to assist the worker in his process of adjustment to the working, home and community environment.
  - It also includes guiding the worker about various legislations available and social security schemes existing for the benefit of workers.

3. Personal Protective Equipment (PPE)

- To prevent the penetration of the uncontrolled hazardous exposure in the body e.g. masks for dust or gases exposure, earmuffs of earplugs for noise exposure, leaded aprons for radiation exposure etc.
  - Note: protective devices must be suitable, comfortable, don't interfere with job continuity and periodically maintained otherwise it will not be used

- **4. Legislative Measures**
- In order to safeguard the health, safety and welfare of the industrial workers, laws have been framed in every country. E.g. new Egyptian Labor Law (2003)

Components of Occupational Health Services:

- **Medical**: carried by occupational physician(s), nurse(s), and technician(s).

  - Pre-employment medical examination.
  - Periodic medical examination.
  - First aid and management of emergencies.
  - Treatment of common diseases and causalities.
  - Health Education.
  - Care of workers nutrition.
  - Records and statistics.
  - Control of communicable diseases.

➢ **Environmental**: Carried by occupational hygienist(s) (engineer or chemist).

- Periodic monitoring of different types of exposure (e.g. Air sampling, noise measurement) and assure that exposure at the work places don't exceed the allowable levels or concentration.
- Advice how to ameliorate the working condition as illumination - ventilation - housekeeping etc….

➢ Social worker specialized in work sociology science:

- Advice how to adopt and ameliorate the interpersonal relation-ship to be cooperative rather than competitive.
- In developed countries psychological work stress is now the most important occupational exposure.

THANK YOU